Practical -2

Problem Definition:

• Extend the portfolio application from Practical 1 by adding React Router.

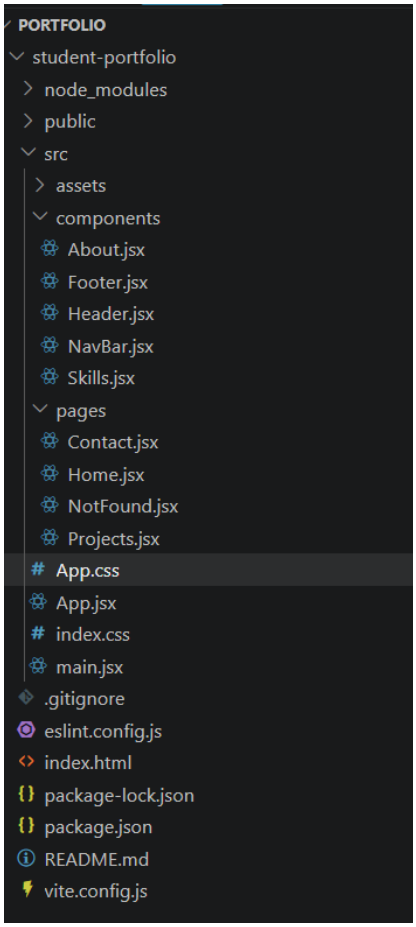
• Implement a navigation bar with at least 3 routes: Home, Projects, and Contact.

• Each route must render a distinct component without a full page reload.

• Add at least 2 useState variables used meaningfully one for toggling UI visibility and one for managing a form input on the Contact page.

• The Contact page must include a controlled input that captures and displays user input in real time.

Folder Structure:



Steps:

1. **Install React Router in the existing portfolio project: npm install react-router-dom**

```
PS C:\Users\Bhumi> npm install react-router-dom
```

2. **Wrap the App component with BrowserRouter in main.jsx: import { BrowserRouter } from 'react-router-dom';**

main.jsx

```jsx
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import { BrowserRouter } from 'react-router-dom'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </StrictMode>,
)
```

3. **Define 3 routes inside App.jsx using Routes and Route: } /> } /> } />**

App.jsx

```jsx
import { useState, useEffect } from 'react'
import { Routes, Route } from 'react-router-dom'
import NavBar from './components/NavBar'
import Home from './pages/Home'
import Projects from './pages/Projects'
import Contact from './pages/Contact'
import NotFound from './pages/NotFound'
import './App.css'

function App() {
  const [darkMode, setDarkMode] = useState(true)

  useEffect(() => {
    document.documentElement.className = darkMode ? 'dark' : 'light'
  }, [darkMode])

  return (
    <div className="app-shell">
      <NavBar darkMode={darkMode} setDarkMode={setDarkMode} />
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/projects" element={<Projects />} />
        <Route path="/contact" element={<Contact />} />
        <Route path="*" element={<NotFound />} />
      </Routes>
```

```
    </div>
  )
}

export default App
```

4. Build the NavBar using Link (not tags) to avoid full page reloads.

NavBar.jsx

```
import { NavLink, Link } from 'react-router-dom'

function NavBar({ darkMode, setDarkMode }) {
  return (
    <nav className="navbar">
      <div className="navbar-brand">
        <span className="brand-mark">BD</span>
        <span>Portfolio</span>
      </div>

      <div className="navbar-links">
        <NavLink to="/" end className={({ isActive }) => isActive ? 'active' : ''}>Home</NavLink>
        <NavLink to="/projects" className={({ isActive }) => isActive ? 'active' : ''}>Projects</NavLink>
        <a href="/#skills">Skills</a>
        <NavLink to="/contact" className={({ isActive }) => isActive ? 'active' : ''}>Contact</NavLink>
      </div>

      <div className="navbar-actions">
        <button className="theme-toggle" onClick={() => setDarkMode(!darkMode)}>
          {darkMode ? '☀' : '🌙'}
        </button>
        <Link to="/contact" className="navbar-cta">Hire Me</Link>
      </div>
    </nav>
  )
}

export default NavBar
```

5. Inside Contact.jsx, create a controlled input tied to useState: const [message, setMessage] = useState(''); setMessage(e.target.value)} />

Contact.jsx

```
import { useState } from 'react'
```

```
3.
4. function Contact() {
5.   const [name, setName] = useState('')
6.   const [email, setEmail] = useState('')
7.   const [message, setMessage] = useState('')
8.   const maxLength = 250
9.
10.  const handleSubmit = (e) => {
11.    e.preventDefault()
12.    alert(`Thanks ${name || 'there'}! Your message has been noted.`)
13.  }
14.
15.  return (
16.    <div className="page">
17.      <div className="bento-card span-3 contact-card">
18.        <h2 className="card-title">Contact Me</h2>
19.        <form className="contact-form" onSubmit={handleSubmit}>
20.          <label>
21.            Name
22.            <input type="text" value={name} onChange={(e) =>
   setName(e.target.value)} placeholder="Your name" />
23.          </label>
24.          <label>
25.            Email
26.            <input type="email" value={email} onChange={(e) =>
   setEmail(e.target.value)} placeholder="you@example.com" />
27.          </label>
28.          <label>
29.            Message
30.            <textarea
31.              value={message}
32.              maxLength={maxLength}
33.              onChange={(e) => setMessage(e.target.value)}
34.              placeholder="Write your message..."
35.              rows={5}
36.            />
37.          </label>
38.          <p className="char-count">{message.length} / {maxLength}
   characters</p>
39.          {message.length > 0 && (
40.            <div className="live-preview"><strong>Live
   preview:</strong> {message}</div>
41.          )}
42.          <button type="submit" className="btn btn-secondary">Send
   Message</button>
43.        </form>
44.      </div>
45.    </div>
```

```
  )
}

export default Contact
```

Home.jsx

```
import { useState } from 'react'
import Header from '../components/Header'
import About from '../components/About'
import Skills from '../components/Skills'

const skillList = [
  { category: 'Frontend', color: '#ff6a3d', skills: ['React', 'JavaScript', 'HTML', 'CSS'] },
  { category: 'Backend', color: '#1d3557', skills: ['Node.js', 'Express'] },
  { category: 'Database', color: '#2a9d8f', skills: ['MongoDB', 'MySQL'] },
  { category: 'Tools & Practices', color: '#e76f51', skills: ['Git', 'VS Code', 'Vite'] },
]

function Home() {
  const [showFact, setShowFact] = useState(false)

  return (
    <div className="page">
      <Header name="Bhumi" tagline=" I turn ideas into websites." />

      <button className="btn btn-secondary fact-toggle" onClick={() => setShowFact(!showFact)}>
        {showFact ? 'Hide quick fact' : 'Show a quick fact about me'}
      </button>
      {showFact && (
        <p className="quick-fact">
          I'm currently in Semester 5 at CHARUSAT, focused on full-stack web development.
        </p>
      )}

      <div className="bento-grid">
        <About />
        <Skills skillList={skillList} />
      </div>
    </div>
  )
}
```

```
36.
37.export default Home
```

Supplementary Problem:

1. **Add a 404 Not Found route that renders a custom error component for unknown paths.**

NotFound.jsx

```
4. import { Link } from 'react-router-dom'
5.
6. function NotFound() {
7.   return (
8.     <div className="page not-found-page">
9.       <h1>404</h1>
10.      <p className="card-text">Oops — this page doesn't exist.</p>
11.      <Link to="/" className="btn btn-primary">Back to Home</Link>
12.    </div>
13.  )
14.}
15.
16.export default NotFound
```

2. **Store the contact form input in state and display a live character count below the input. in this**

Projects.jsx

```
50.const projects = [
51.  {
52.    id: 1,
53.    title: 'Personal Expense Tracker (UI/UX)',
54.    tech: ['Figma'],
55.    points: [
56.      'Designed user-friendly interface and wireframes',
57.      'Focused on clean and intuitive user experience',
58.    ],
59.  },
60.  {
61.
62.    title: 'Beauties & Bows – E-commerce Website',
63.    tech: ['HTML', 'CSS', 'JavaScript'],
64.    points: [
65.      'Developed a responsive front-end website for a fashion brand',
66.      'Implemented product display and user-friendly navigation',
67.    ],
```

```
  },
  {

    title: 'Note-Taking Web Application',
    tech: ['PHP', 'MySQL'],
    points: [
      'Created a web app to add, edit, and delete notes',
      'Integrated backend with database for data management',
    ],
  },
  {
    title: 'Shake & Bake Website',
    tech: ['WordPress'],
    points: [
      'Developed and customized a business website using WordPress',
      'Managed layout, design, and content',
    ],
  },
]

function Projects() {
  return (
    <div className="page">
      <div className="bento-grid">
        {projects.map((p, index) => (
          <div key={p.id} className="bento-card span-3">
            <p className="card-number">{String(index + 1).padStart(2, '0')}</p>
            <h2 className="card-title">{p.title}</h2>
            <ul className="project-points">
              {p.points.map((point, i) => (
                <li key={i}>{point}</li>
              ))}
                  </ul>
                  <div className="card-tech">
                    {p.tech.map((t) => (
                      <span key={t} className="tech-tag">{t}</span>
                    ))}
                  </div>
                </div>
              ))}
            </div>
          </div>
        )
      }

      export default Projects
```

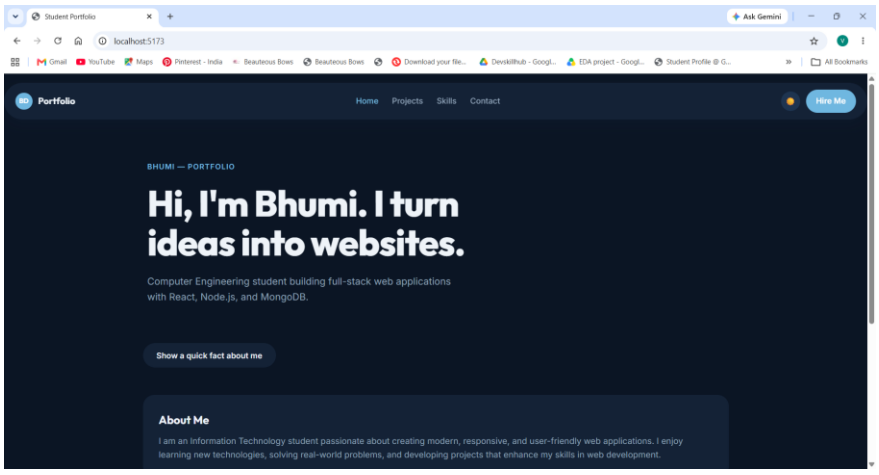
Header.jsx

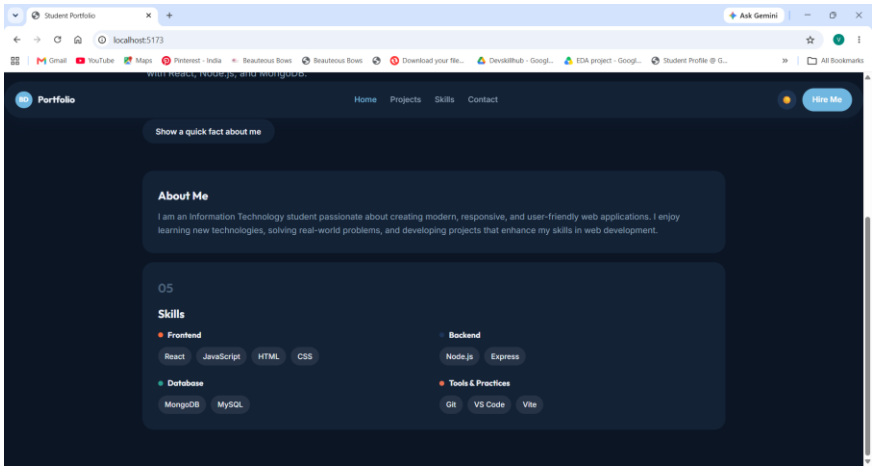
```
function Header({ name, tagline }) {
  return (
    <header className="hero">
      <p className="hero-eyebrow">{name} &mdash; Portfolio</p>
      <h1 className="hero-title">Hi, I'm {name}. {tagline}</h1>
      <p className="hero-subtitle">
        Information Technology student building full-stack web applications
        with React, Node.js, and MongoDB.
      </p>
    </header>
  )
}

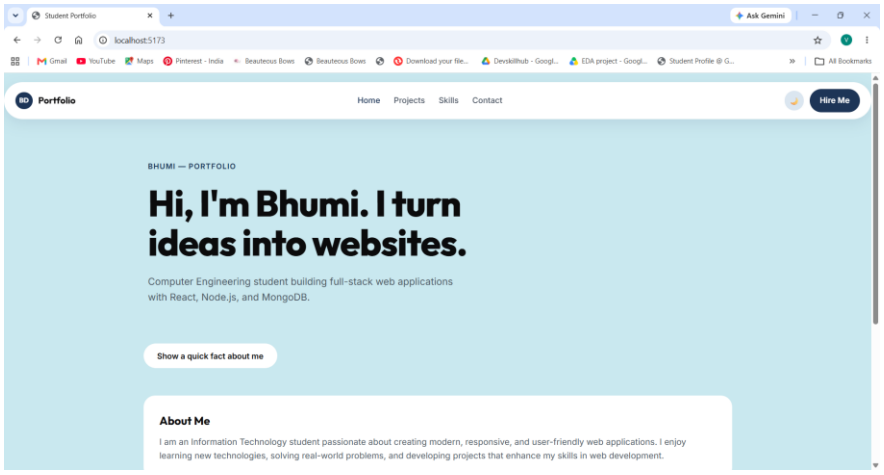
export default Header
```

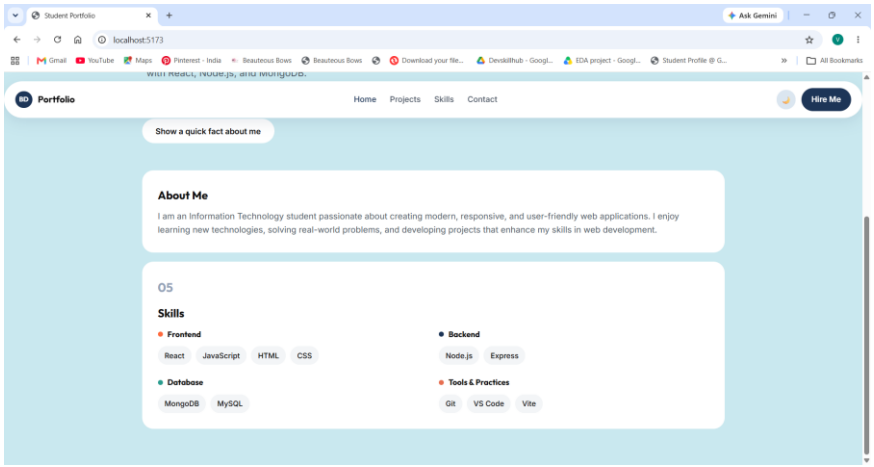
7. Test navigation across all 3 routes and confirm no page reload occurs

Output:






Student Portfolio
localhost:5173
Ask Gemini
Portfolio
Home
Projects
Skills
Contact
Hire Me
BHUMI — PORTFOLIO
Hi, I'm Bhumi. I turn ideas into websites.
Computer Engineering student building full-stack web applications with React, Node.js, and MongoDB.
Show a quick fact about me
About Me
I am an Information Technology student passionate about creating modern, responsive, and user-friendly web applications. I enjoy learning new technologies, solving real-world problems, and developing projects that enhance my skills in web development.


Student Portfolio
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05
Skills
Frontend
React
JavaScript
HTML
CSS
Backend
Node.js
Express
Database
MongoDB
MySQL
Tools & Practices
Git
VS Code
Vite